

A SURVEY ON TABULAR DATA AUGMENTATION FOR EMPLOYEE ATTRITION PREDICTION

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Abstract—The explorations of this paper are directed to the use of data augmentation to improve the accuracy of employee attrition prediction models. While machine learning models are valuable tools, their effectiveness relies heavily on the quality and quantity of training data. Data augmentation offers a solution by artificially increasing the size of datasets with new synthetic data points that resemble real data. After a comprehensive survey on the available augmentation models, the findings reveal that in as much as Generative Adversarial Networks (GANs) are useful for image analysis and Natural Language Processing tasks, it is less effective in augmenting structured data since there is need to maintain the correlations in columns. GANs may get lost in trying to connect the dots. In addition, there is evidence that the use of Easy Data Augmentation (EDA) by randomly deleting or inserting data points and adding small amounts of noise to the data has a low success rate. It does not follow a predetermined procedure and its operations are not explainable since they are random. The paper concludes that WCGAN-GP appears to be the most effective technique for generating statistically sound synthetic data for employee attrition prediction. This is because WCGAN-GP preserves complex correlations between features due to its conditional nature and can produce range bound data due to its gradient penalty nature. Moreover, the generated data can be more difficult to trace back to the original dataset hence insuring data privacy since we are dealing with employee data. However, it suggests that combining WCGAN-GP with XGBoost, an ensemble learning technique, might not be the most optimal approach. But for future research the paper proposes investigations on the potential of combining SMOTE and WCGAN-GP for data augmentation. Since GANs May struggle with data sparsity when there are very few examples of certain combinations of features. SMOTE might help to balance the dataset. This will minimize statistical bias and promote generation of realistic data hence improving the effectiveness of employee attrition prediction models.

Keywords — Data Augmentation, Attrition, WCGAN-GP, SMOTE, GANs.

I. INTRODUCTION

Employee attrition is the gradual reduction in the size of a workforce due to employees leaving the company and not being replaced. It is a natural part of any business, but high levels of attrition can be disruptive and costly [1]. To overcome

the negative impacts of employee attrition, such as reduced productivity and low employee motivation corporates need to anticipate attrition and be better prepared for it. This can be made possible through employee attrition prediction using machine learning models. However, machine learning models are not 100% accurate because they learn from real world data which is sometimes limited or even Biased [2].

The predictive models use a systematic and data-driven approach. This typically involves the use of historical data and statistical models to project future outcomes. Their accuracy is directly proportional to the availability of Larger Training Datasets [3]. This is the main discussion in this survey. Improving the accuracy of prediction models by augmenting employee data, which involves creating new synthetic data points from the existing without altering the statistical parameters of the dataset or introducing bias [4]. The main motive behind is because employee attrition data is expensive to collect and is collected for longer time periods than they need to be used.

II. LITERATURE REVIEW

A. Related Work on Data Augmentation Techniques

The aim is to improve the accuracy of employee attrition prediction through Data Augmentation, a technique that can be used to artificially increase the size of a dataset by creating new data samples that are almost indistinguishable to the real data. This means that the data should have similar statistical parameters to the original data even though it might not be identical to the real data, that the purpose anyway, to ensure diversity and give machine learning algorithms variety of data to learn from [5]. While we frequently discuss augmentation in the context of images or text, tabular data holds its unique challenges and opportunities. While challenges exist, the potential benefits for model performance and robustness are significant [6]. The industry has since savoured hybrid techniques which have turned out to be lucrative solutions in Tabular Data Augmentation. As with all techniques it is crucial to understand our data, experiment responsibly and validate thoroughly [7]. To gain a comprehensive understanding of the current research landscape in Data Augmentation this review critically analyses various scholarly articles [8]–[14] and identifies key findings and research trends with emphasis on methodologies used, how they fared and their challenges. The nature of the review on the listed papers is both empirical and technical, giving much attention to the technicalities and results of each paper.

B. Generative adversarial Networks (GANs)

GANs are a type of deep learning model that can generate new data samples indistinguishable from the training data. GANs work by training two networks simultaneously: a generator network that creates new data samples and a discriminator network that distinguishes between real and generated data.[8]. Improvement to the generator network is established by attempting to deceive the discriminator network, while the discriminator network improves by accurately classifying real and generated data. The training process is iterative until an equilibrium is reached between the adversarial loss of the generator and the discriminator. In the end the Generator is able to generate data similar to the training data. The generator network can then produce new images that are similar to the training data. Fig.1 shows the two adversarial networks are trained together. The generator is trained to generate new realistic data that are indistinguishable from real data, while the discriminator determines whether the data are real or generated [8].

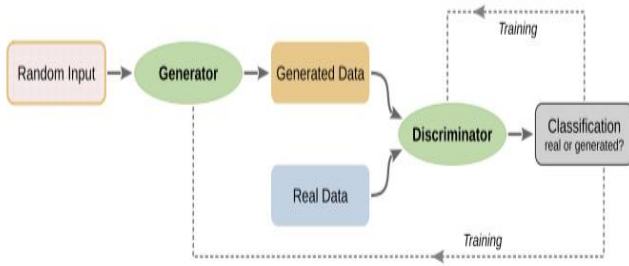


Figure 1 : GAN Architecture[8]

GANs are less commonly used for generating tabular data. One reason for this is that generating tabular data requires a more structured approach than generating images or natural language [15]. GANs are great at creating images and writing, but when it comes to tabular data, things get a bit messy. In tabular, data points aren't random, they are correlated. We need the generated data to reflect these connections, like mimicking the spread of values in a column or the way different columns interact. Imagine a table about weather, sunny days are usually warmer. The generated data should show this pattern too. Some data is like a ruler; it has a range of values. Other data is like picking colours from a palette - it has limited options. GANs struggle with these limited choices, making it tricky to capture the variety within a category (like all the different types of weather we can have on a sunny day). If some combinations of data rarely appear in the original table, the GAN might get lost trying to dream them up[16]. This can lead to strange or nonsensical results in the generated data. Despite these challenges, researchers are making progress! New methods are being developed to address these issues and improve the quality of GAN-generated tables. While it's still an ongoing area of research, GANs have the potential to become a powerful tool for creating realistic and informative tabular that could potentially achieve robustness and better generalisation in training machine learning models hence improved accuracy [17].

C. Generative Adversarial Network modelling inspired from Naive Bayes and Logistic Regression's relationship (GANBLR)

[9] Acknowledges the state of the art models of tabular data generation which are based on the base Model GAN, such as CTGAN, TableGAN, MedGAN etc. However, these models have some weaknesses since they only focus on the performance of the model and limited emphasises given to the interpretation of the model and they only operate on raw features only and hence fail to exploit any prior knowledge on explicit feature interactions that can be utilized during the data generation process [8]. To the limitations, the author proposed a novel tabular data generation model – Generative Adversarial Network modelling inspired from Naive Bayes and Logistic Regression's relationship (GANBLR), which can not only address the interpretation limitation in existing tabular GAN-based models but can provide capability to handle explicit feature interactions. GANBLR which uses BN^e (Bayesian Network as the Generator) as the generator and BN^d (Bayesian Network as the Discriminator) as the discriminator. The generator in GANBLR cooperates with the discriminator to form an adversarial structure to improve the quality of synthetic tabular data. The model is only limited to producing datasets that have categorical attributes only. BN^e has been studied in the context of classification this is the first work which studies its effectiveness as a data generation technique. The Authors Compared GANBLR to existing SOTA GAN models on 15 public tabular datasets. The results demonstrate that GANBLR outperforms in the machine learning utility and it also provides better interpretability. Model interpretability refers to how easy it is for humans to understand the reasoning behind a model's predictions. In other words, it's about understanding why a model makes the decisions it does. [18].The proposed augmentation Technique also produces synthetic data that is statistically consistent with the original. This notion is measured with Jensen-Shannon Divergence and Wasserstein Distance[9]. Fig. 2 Shows an architecture of the GANBLR model with the Bayesian Network as the Generator and Bayesian Network as the Discriminator.

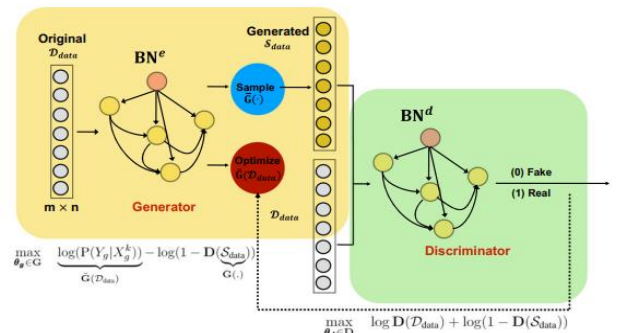


Figure 2 : GANBLR Architecture[19]

D. Wasserstein Distance Generative Adversarial Network vs SMOTE Density-based Over-sampling

Report [10] postils that Simple Minority Oversampling Technique (SMOTE) pioneered as a solution to imbalanced datasets and to improve accuracy in Binary classification.

Alternatively, Generative Adversarial Networks (GANs) can perform the same function. Not only do GANs perform this faster but it also ensures accuracy by addressing the problem with SMOTE which is misappropriation of the true underlying probability of the minority class data distribution. GANs can sample from the true underlying probability without explicitly specifying its form.

Some arguments have the opinion that SMOTE is prone to subjectivity due to the randomness of its operations [20]. Taking in to consideration a Random instance “x”, found in the minority class. To generate a new data point “s” SMOTE will calculate the distances between this “x” and all the other data points using a method called Euclidean distance. This tells us which data points are most similar (closest) to “x”. The similar points are called its k-Nearest Neighbours (k-NNs). One of these k-NNs, it will randomly choose one. This random chosen neighbour becomes the basis for our new data point. It is labelled “s” as a new sample. The mathematical equation (1) summarizes the generation of a new instance in the minority class or in the majority in the event that SMOTE used as an augmentation tool [21].

$$(s = x + \alpha(y - x)) \quad (1)$$

Comparison was on the two methods for generating synthetic data, Wasserstein Conditional GAN with Gradient Penalty (WCGAN-GP) and a SMOTE density-based approach on multiple datasets using metrics like precision, recall, and F1-score and a logistic regression algorithm. The results show a significant potential for GANs as an alternative to SMOTE density techniques, useful for new sample creation, data augmentation and boosting classification models. The conclusion implies that While SMOTE is a useful technique, it doesn't strictly adhere to the original data's underlying probability distribution. This can lead to subjective and potentially inaccurate synthetic data, especially when the goal is to precisely replicate the statistical properties of the real data.

E. Received Signal Strength Indication (RSSI)

Rashmi S. Sinha and Seung H Hwang introduces a new method to improve indoor positioning systems by synthetic data creation. It uses signal strength readings (RSSI) from a single location to create more realistic variations, mimicking how these signals naturally fluctuate in real-world environments [11]. Existing methods don't account for the constant change, which can limit their accuracy hence RSSI was such an improvement. Similar to adding random noise to signal strength readings. The authors introduced controlled noise to numerical values in the tabular data. In addition, they applied Scaling and shifting inspired by adjusting signal strength. Scaling is multiplying by a constant and shifting adding a constant to values in specific columns of data creating new data points with slightly different magnitudes or baseline potentially reflecting real-world variations. This method might produce data points that are statistically inconsistent with real data as the constants used for shifting and scaling are prone

subjectivity. However, the technique achieved high test accuracy and low average location error in both laboratory simulation and real time environment. In addition, the absence of techniques or methods that are used to measure the statistical similarity of the synthetic data to the original data poses weakness if RSSI was to be chosen as a data augmentation technique since we won't have a way to validate the generated data points for statistical bias.

F. Back Translation (BT) and Easy Data Augmentation (EDA)

The authors in [12], In order to improve the accuracy of their classification model, they employed two data augmentation methods, Back Translation and Easy Data Augmentation which creates a richer training dataset. Back translation isn't directly applicable to tabular data due to its inherent structure. Tabular data consists of rows and columns representing specific features and their values. Back translation relies on translating text from one language to another and back, introducing variations that wouldn't be meaningful in a tabular format. Values in tabular data represent concrete information like numbers, categories, or dates. Translating them wouldn't introduce new or meaningful variations. "Age: 30" translated to another language and back might return "Age: 30" (in the target language) hence no new information is gained. Tabular data relies on a structured organization. Back translation shuffles the order of words in sentences, which wouldn't translate well to the row-column structure of tables [22].

On the other hand, the results indicate that the implementation of the EDA techniques successfully improved the performance of the classifiers. EDA was implemented by randomly swapping or shuffling values within rows or columns of tabular data, randomly deleting or inserting data points and adding small amounts of random noise to numerical values in tabular data. The processes in EDA will successfully increase the variety of data points within the dataset but may not always introduce meaningful variations since it does not follow a predetermined procedure and its operations cannot be explained since they are random [12].

G. Wasserstein Conditional Generative Adversarial Network (WCGAN-GP)

Walia et al [13] explores whether a technique called WCGAN-GP can create synthetic data for tables that closely resembles real data, without accidentally revealing any sensitive information and losing some important correlations. The authors had aimed to address privacy concerns that often limit the use of valuable data. Synthetic data offers a secure solution, enabling rapid and cost-effective innovation. WCGAN-GP belongs to the broader category of Generative Adversarial Networks (GANs) which has a pair of competing neural networks, the generator which aims to master the implicit distribution of the real data and generate new data points that resemble the real data as closely as possible whilst the discriminator tries to distinguish between the real data and the generated data by the generator. WCGAN-GP has an added concept of Wasserstein Condition and Gradient penalty which

governs the quality of data being generated that it stays within the parameters of real world Data [23].

It is important to appreciate the other base models that make up the WCGAN –GP model because in a way the technique is a hybrid of other GAN techniques. Firstly, we have the Conditional GAN (CGAN) which incorporates “an extra” class of labels by both the Discriminator and the Generator Network [24]. For example, to generate synthetic data points for an age column we will have to specify a threshold based on the average life expectancy of an average human and the legal retirement age. In this case this will be our Condition e.g. 16 – 65, will make sure the variations of synthetic data points fall within that range. However, arguments suggest that the techniques’ performance cannot be generalised over different column data types since it has only proven to be useful in numerical Data types [25]. According to [13] the second variation is the Wasserstein GAN(WGAN) which will transcode to WCGAN when combined with Conditional GANs. WGANs introduce the Wasserstein loss, a metric that provides a more meaningful measure of distance between real and generated data and the discriminator is actually called a critic instead of a discriminator. The critic uses a scoring system instead of a simple classification. Real samples get high scores, while fakes get low scores. The Wasserstein loss then measures the distance between the probability distributions of synthetic and real samples.

Lastly, there is the WGAN with Gradient Penalty WGAN-GP which transcode to WCGAN-GP when the CGAN is incorporated. A key innovation of WGAN-GP is the gradient penalty. This penalty term is added to the critic's loss function. It helps to enforce a specific constraint on the gradients of the critic, leading to more stable training and preventing issues like vanishing gradients that can plague GAN training. Instead of clipping the weights of the critic directly, WCGAN-GP takes a different approach. It introduces a penalty term called the gradient penalty. This penalty encourages the critic to have gradients with a specific property: a norm of 1 (think of norm as the magnitude). By enforcing this property, WCGAN-GP makes the critic's training more stable and prevents its gradients from exploding [26].

The performance of WCGAN-GP is evaluated against SMOTE thrice with realistic datasets from different domains WCGAN-GP proved to be a powerful tool for creating synthetic tabular data. The generated data captured the original data's patterns, distributions, and relationships very well. Additionally, it outperformed the traditional SMOTE method in two ways, it improved the effectiveness of machine learning models trained on the data and it offered better protection for the privacy of the original data.

H. WCGAN-GP and XGBoost Hybrid

Finally, paper [14] argues that a hybrid of Wasserstein Conditional Generative Adversarial Network with Gradient Penalty (WCGAN-GP) and an XGBoost Classifier can significantly improve intrusion detections. The proposed model is evaluated with other standard GAN models and it

reflected better results. XGBoost (Extreme Gradient Boosting) is primarily not considered a data augmentation tool in the traditional sense. Its core functionality lies in ensemble learning and boosting to improve the performance of machine learning models, particularly for regression and classification tasks. However, XGBoost can indirectly contribute to data augmentation through its ability to handle missing values and feature engineering since it is harder to implement Generative adversarial networks on data with missing values. Not only are the augmentation results impressive but they also spearhead a new way of thinking that prompts the combination of different techniques for data Augmentation.

III. CONCLUSION AND PROPOSED FUTURE WORK

The reviewed studies [10]–[14] had a universal objective, improving the size of the datasets so as to improve the performance of prediction models. The findings in the paper suggest that Generative Adversarial Networks (GANs) are useful for image analysis and Natural Language Processing tasks, it is less effective in augmenting structured data since there is need to maintain the correlations in columns. GANs may get lost in trying to connect the dots. In addition, there is evidence that the use of Easy Data Augmentation (EDA) by randomly deleting or inserting data points and adding small amounts of noise to the data has a low success rate in generating synthetic data that is indistinguishable from the real data since it does not follow a predetermined procedure and its operations cannot be explained since they are random [12].

The paper concludes that WCGAN-GP appears to be the most effective technique for generating statistically sound synthetic data for employee attrition prediction. This is because WCGAN-GP preserves complex correlations between features due to its conditional nature and can produce range bound data due to its gradient penalty nature. Moreover, the generated data can be more difficult to trace back to the original dataset hence insuring data privacy since we are dealing with employee data. Lastly, evidence point that combining WCGAN-GP with XGBoost, an ensemble learning technique, might not be the most optimal approach although it can indirectly contribute to data augmentation through its ability to handle missing values it was originally meant for ensemble learning and Boosting.

Finally, for future research the paper proposes investigations on the potential of combining SMOTE and WCGAN-GP for data augmentation. Since GANs May struggle with data sparsity when there are very few examples of certain combinations of features. SMOTE might help to balance the dataset. This will minimize statistical bias and promote generation of realistic data hence improving the effectiveness of employee attrition prediction models [10].

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